

Reactive Visuals in P5.js with Custom Analog and Digital Inputs

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ABSTRACT

This SIGGRAPH lab is an introduction to creating reactive graphics in p5.js that respond to inputs from external sensors connected through a hand-built circuit via the Raspberry Pi Pico microcontroller. Participants are guided through the process of building a working hardware and software template that can be further customised for their own creative designs.

KEYWORDS

analog and digital sensors, arduino, interface, p5js, physical computing, processing, prototyping, raspberry pi pico, reactive visuals

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1 INTRODUCTION

This lab session is an introduction to creating reactive graphics in p5.js that respond to external sensor inputs from a custom built circuit connected to a Raspberry Pi Pico. P5.js is a versatile open-source JavaScript library with a browser-based IDE, ideally suited for programming interactive visuals [Ye and Lee McCarthy 2023]. The Pi Pico is a powerful low-cost microcontroller programmable in MicroPython and also compatible with the Arduino IDE.

This introductory level project leverages both mediums to create animated abstract visuals that respond to external analog and digital inputs from the attached hardware prototype.

All participants will be guided through setting up the necessary coding environments, software, and hardware circuit (Figure 1). The class cohort will also be shown how to customise the graphics and animations in the given code example, referring to best-practice examples of graphics customisation in p5.js.

2 PREREQUISITES

A computer running Windows, Linux, or macOS and also:

- An up to date web browser.
- A user account at p5js.org.
- P5.serialcontrol.

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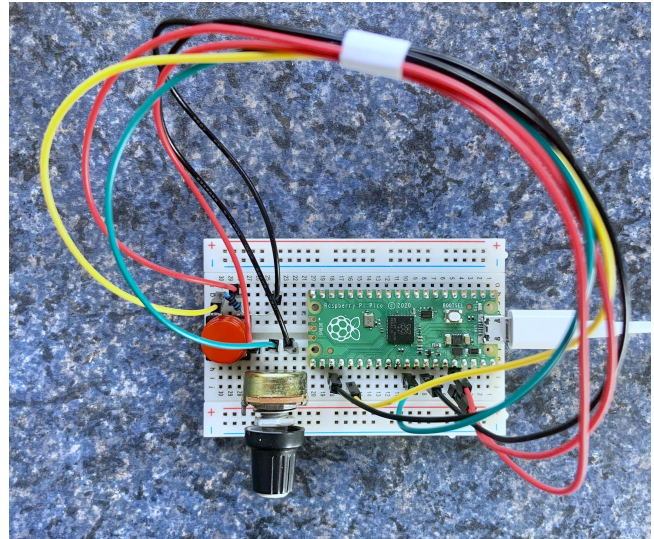


Figure 1: A Raspberry Pi Pico powered custom breadboard circuit with analog and digital inputs.

Pi Pico Microcontroller (with headers) and a USB Cable.
Half-Size Solderless Breadboard.
Resistor (x1, 10K Ohm value).
Pushbutton (x1).
Potentiometer (x1).
Jumper Wires (x6).

3 ONLINE RESOURCES

Up to date source files and references for this tutorial are provided at github.com/kierannolan and kierannolan.com.

4 HARDWARE

4.1 Raspberry Pi Pico

Introduced in 2020, the Pi Pico is a versatile and low cost microcontroller [Raspberry Pi Ltd 2023]. It has many possible uses, including connecting custom analog and digital inputs and outputs to a host computer through its GPIO (General Purpose Input/Output) pins.

The Pi Pico W was released in 2022 [The Raspberry Pi Foundation 2023], it has the same form factor and pin layout as the original Pi Pico but also features built-in Wi-Fi and Bluetooth.

4.2 Solderless Breadboard

A prototyping breadboard allows quick and convenient prototyping of circuits without the use of soldering. It contains spring loaded

contacts that connect wires in horizontal and vertical lines depending on where they're placed on the board. This design allows wires and components to be easily moved around.

4.3 Analog Sensors

Analog sensors send a range of values. Everyday examples of analog sensors include volume controls and light dimmers. They all connect the same way to the circuit using 3 wires: voltage in (+3.3 volts on the Pi Pico), signal out (to an analog pin such as A0, analog zero), and ground (GND). The Pi Pico has 3 analog input pins.

This project uses a potentiometer, but it can be easily replaced with a different type of analog sensor, such as a pressure sensor, temperature sensor, distance sensor, flex sensor, or one of the many other types available.

4.4 Digital IO

26 of the pins on the Pi Pico are multipurpose GPIO pins, meaning these pins can be set as either digital inputs or digital outputs. Digital inputs allow connecting up of a switch that inputs a value of either on or off. There are many varying types of push button switches, including magnetic switches and motion activated tilt switches.

On is referred to in the code as 'high', and off is 'low'. These values translate as 1 for on or high, and 0 for off or low. Digital outputs can switch on or off devices that move between two voltage states, such as LED lights, solenoids (electromagnetic actuators), and optoisolators (electronic switches).

5 SOFTWARE

5.1 Pi Pico Microcontroller Code

The microcontroller code for this project is available as a pre-prepared .uf2 file from the project's GitHub [Nolan 2023] page (see references section). This file was created using the Arduino IDE [Arduino 2023]. Plug the Pi Pico into your computer while holding down the button on the Pico. Release the button on the Pico. The Pico will show up as a USB drive. Copy the .uf2 file over to the Pico drive.

The Pico will disappear from your list of hard drives, meaning it's updated with the new code and is working. You don't have to repeat this process again, the microcontroller code will remain on the Pico when it is powered off.

5.2 P5.serialcontrol and p5.js Source Code

P5.serialcontrol [js6450 2019] acts as a bridge between the Raspberry Pi Pico hardware connected to the USB port and p5.js running in the web browser. It passes data between both, allowing p5.js to read in signals from sensors and switches connected to the Pi Pico, and also allowing control of external outputs such as LEDs and servos on the microcontroller from p5.js.

Make sure that the COM port details in your sketch.js file matches the COM port as detected in the p5.serialcontrol application. The p5.js source code is available on this project's GitHub [Nolan 2023] page

Also reference the p5.serialport.js library in the index.html file of your project in the p5.js Web Editor according to the sample code.

6 CIRCUIT ASSEMBLY

If you're using a Pi Pico with the pins soldered on, plug the legs directly into rows B and G. The end with the USB connector should align with row 1 at the top of the board.

6.1 Breadboard Wiring Instructions

Add the jumper wires to the breadboard as follows:

- Red Wire 1: I5 to B28.
- Red Wire 2: J5 to F26.
- Black Wire 1: J8 to F22.
- Black Wire 2: J18 to B24.
- Green Wire: J10 to F24.
- Yellow Wire: J16 to B30.
- Resistor: C24 to A30.
- Potentiometer: J22, J24, J26.
- Pushbutton: D28, D30, G28, G30.

7 SUMMARY

Install the p5.serialcontrol application. Update the Pi Pico by copying the .uf2 file to it while it is in USB storage mode (activated by holding the Pico button for a moment when plugging in the drive). Log into editor.p5.js and transcribe the supplied p5.js code, also make sure to edit index.html as instructed.

Start up p5.serialcontrol, and click 'rescan ports', taking note of the ports listed. Connect your custom Pi Pico breadboard circuit through USB and run p5.serialcontrol again, select the new port that has appeared and click 'Open'.

Select 'console enable' and 'read in ASCII' in the p5.serialcontrol control panel. The values from the Pi Pico sensors will display in the window. This will be a two numbers separated by a comma. The first is the value from the analog sensor, and will range from 1023 to 4. The second number is the input from the digital button which will trigger a 1. When the button isn't pressed the second digit will remain at zero. Edit the p5.js code so that the COM port number matches the one selected in p5.serialcontrol.

Run the p5.js sketch button, then turn the potentiometer dial and push the button on your prototype circuit, while viewing the on-screen results in p5.js. Duplicate your p5.js sketch so that you have a backup of the original, and customise the visuals and animated responses into your own original designs.

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